

01_COGNITIVE_MEMORY_TOPOLOGY_AND_RECONCILIATION.md

Unified Cognitive-Engineering Architecture & 5+1 Tier Memory Model

1. Executive Summary & Reconciliation

Historical drafts explored both a 4-tier model (collapsing procedural logic and run traces into `/automation_scripts/`) and a 5-tier model (separating raw sources, wiki, and JSONL caches, but missing procedural prompt definitions).

The unified canonical architecture resolves this divergence by establishing a strict **5+1 Tier Cognitive Memory Model** that maps human cognitive functions directly to deterministic AI engineering structures:

Tier	Canonical Directory	Cognitive Memory Equivalence	Primary Function & Artifact Types
Tier 1	<code>01_raw_sources/</code>	Sensory / External Ground Truth	Immutable PDFs, technical manuals, transcripts, media dumps. Read-only.
Tier 2	<code>02_wiki_md/</code>	Semantic Memory (Conceptual)	Human-readable Markdown linked graph, architectural blueprints, entity registries.
Tier 3	<code>03_recall_cache/</code>	Working Memory Accelerator	Line-delimited JSONL (<code>chunk.jsonl</code>), vector indexes, and low-latency KV tables.
Tier 4	<code>04_skills_runtime/</code>	Procedural Memory (Skills & Habits)	Executable scripts (<code>.py</code> , <code>.sh</code>), prompt

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			templates (SKILL.md), runtime policies.
Tier 5	05_episodic_logs /	Episodic Memory (Trajectories)	Time-indexed run traces, RLVR assertion outcomes (+1.0 / -1.0), quarantine logs.
Root	MAP.md & manifest.jsonl	Metacognitive Index & Navigation	Global ontology graph, cryptographic hash registries, and cross-tier routing pointers.

2. Human Cognitive Memory vs. AI Engineering Memory in KAG/RLVR

Memory Type	Human Cognitive Function	AI System Engineering Counterpart	Operational Role in KAG Recurse & RLVR Training
Episodic	Personal History: Remembering specific past events, times, and errors.	Execution Logs & Vector Stores (05_episodic_logs/)	Logs the exact sequence of an attempt (Prompt → Tool Call → Error Trace → Verifier Score).
Procedural	Habits & Skills: Automated, unconscious execution routines.	Algorithmic Policies & Model Weights (04_skills_runtime/)	Permanent behavioral intuition developed via RLVR to backtrack and self-correct on error states.

Memory Type	Human Cognitive Function	AI System Engineering Counterpart	Operational Role in KAG Recurse & RLVR Training
Semantic	Facts & Concepts: Generalized objective knowledge.	Static Parametric Weights & Living Wiki (02_wiki_md/)	Foundational domain concepts, programming syntax, and architectural specs.
Working / Short-Term	Mental Workbench: Holding 4–7 active items in immediate awareness.	The Context Window & Fast Cache (03_recall_cache/)	Active prompt system instructions, injected user habits, and immediate task branch variables.

3. The Continuous Evolution Flywheel

In recursive training and online agent adaptation, these memory layers interact continuously:

1. **Ingress:** The agent executes a step utilizing its Working Memory (Context Window loaded via [03_recall_cache/](#)).
2. **Telemetry:** Every tool call and outcome is logged into Episodic Memory ([05_episodic_logs/](#)).
3. **Verification:** The RLVR verifier evaluates episodic traces, assigning rewards (+1.0) or penalties (-1.0).
4. **Refinement:** Audited successes update Procedural prompt skills and fine-tuning datasets, permanently upgrading baseline performance.